

Notes: Bisetti & Karolyi (Journal of Finance, 2024)

Meeting Targets in Competitive Product Markets

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1 Big picture

- **Question.** How do public firms change consumer product prices when managers face short-term pressure to meet EPS targets?
- **Setting.** The paper uses U.S. commercial banks and retail deposit products, where prices are observable as deposit APYs at the bank-product-county-quarter level.
- **First result.** Banks that barely miss EPS targets experience a negative announcement-return notch of roughly 58 basis points.
- **Bunching result.** The EPS surprise distribution shows abnormal mass just above zero.
- **Main pricing result.** Banks in the estimated spike range cut deposit rates by about 1.8 to 2.5 basis points across specifications.
- **Mechanism.** Cutting deposit rates lowers interest expense, raises current EPS, and helps some banks avoid the negative return notch.
- **Cost.** Rate cuts trigger deposit outflows and franchise-value losses.
- **Competition.** Local competitors, including private banks, raise deposit rates when nearby public banks target earnings.
- **Economic takeaway.** Performance targeting affects consumer prices, but product-market competition partially offsets the targeting bank’s attempt to lower depositor compensation.

2 Data and measurement

2.1 EPS targets and EPS surprises

The EPS target for bank i in quarter t is the median analyst consensus EPS forecast using analysts’ most recent forecasts 10 to 30 weeks before the earnings announcement. EPS surprise is measured as a percentage difference between realized EPS and target EPS. The main EPS sample focuses on observations around the zero-surprise threshold.

Interpretation. The target definition is designed to capture targets managers can plausibly observe and respond to before the end of the quarter.

2.2 Deposit-rate data

The paper uses RateWatch APYs for money market accounts, savings, checking, and 3-, 6-, 12-, 24-, and 36-month CDs. APYs are aggregated to bank-county-quarter averages for each product. Short-maturity products include MMAs, savings, checking, and 3-month CDs; long-maturity products include CDs of 6 months or longer.

Interpretation. Deposit rates are a clean consumer-price setting because the same bank can offer standardized products across counties and over time.

2.3 In Spike indicator

$$InSpike_{it} = \begin{cases} 1, & \tilde{y}_{it} \in [\hat{y}_{Low}, \hat{y}_{High}], \\ 0, & \text{otherwise.} \end{cases}$$

Bank-quarter observations in the spike range are interpreted as likely targeting observations.

Interpretation. In Spike connects the bunching evidence to deposit-rate behavior.

3 Models and empirical specifications

3.1 Bunching estimation

$$\hat{h}(j) = n_j/d, \quad B = \int_{\hat{y}_{Low}}^{\hat{y}_{High}} \left(\hat{h}(\tilde{y}) - \hat{h}_0(\tilde{y}) \right) d\tilde{y}, \quad -\Delta\hat{y} = -\frac{B}{\hat{h}_0(0)}.$$

Interpretation. The estimator compares the observed EPS surprise density near zero to a smooth counterfactual density.

3.2 Conceptual model of earnings management

$$y = y^* + q, \quad R(\tilde{y}^*, \tilde{q}) = \tilde{y}^* + \delta\tilde{q}, \quad \delta = \frac{(1-\lambda)r}{1+r}.$$

Managers choose earnings management to maximize

$$V(R, \tilde{q}) = R(\tilde{y}^*, \tilde{q}) - g(\tilde{q}), \quad R_N(\tilde{y}^*, \tilde{q}, \tilde{y}) = \tilde{y}^* + \delta\tilde{q} - \xi \mathbf{1}\{\tilde{y} < 0\}.$$

Interpretation. The return notch creates an incentive for banks near zero to increase earnings management and avoid missing the target.

3.3 Deposit-rate regression

$$d_{ipct} = \alpha + \beta InSpike_{it} + \gamma_{FE} + \varepsilon_{ipct}.$$

Interpretation. β measures whether targeting banks offer lower APYs for the same product and county during targeting quarters.

3.4 Deposit quantity regression

$$D_{it} = \alpha + \beta InSpike_{it} + \gamma_{FE} + \varepsilon_{it}.$$

Interpretation. A negative β for deposits indicates franchise costs from rate cutting.

3.5 Event-study design

$$d_{iop\tau} = \alpha + \beta_1 MoFE_{o\tau} + \beta_2 InSpike_{io} \times MoFE_{o\tau} + \gamma_{FE} + \varepsilon_{iop\tau}.$$

Interpretation. This design checks whether rate cuts begin in the targeting quarter and reverse afterward.

3.6 Competitive spillover regression

$$d_{ipct} = \alpha + \beta_1 InSpike_{it} + \beta_2 InSpike_{it} \times \%InSpike_{ct} + \beta_3 (1 - InSpike_{it}) \times \%InSpike_{ct} + \beta_4 Public_{it} + \gamma_{FE} + \varepsilon_{ipct}.$$

Interpretation. β_3 captures whether nontargeting banks raise rates when more local competitors have targeting incentives.

4 Identification and empirical design

- Figure 1 establishes that missing the EPS target creates a market penalty.
- Figure 2 and Table II identify the spike range where EPS surprises appear manipulated toward zero.
- Tables III and IV connect this spike range to deposit-rate cuts with fixed effects and robustness tests.
- Table V validates that rate cuts flow through to bank-level deposit interest expense.
- Figure 3 rules out pre-trends and shows temporary targeting-quarter cuts.
- Table VII shows deposit outflows, measuring the cost side of targeting.
- Tables VIII and IX show competitive responses by local nontargeting banks.

5 Tables and figures with expanded interpretation

5.1 Table I: Summary statistics

Read: EPS surprises average 1.41%, average APY is 0.85%, and short-maturity products have much lower average rates than long-maturity products.

Interpretation. Small basis-point cuts are meaningful because baseline deposit rates are low, especially on short-maturity products.

Economic message. Table I anchors the economic magnitudes of later pricing effects.

Table I
Summary Statistics

This table provides summary statistics for the main variables used in the paper. EPS surprise is the percentage difference between realized EPS in quarter t and the median forecast across all analysts' most recent forecasts over the weeks $[-30, -10]$ relative to the actuals' release date. EPS estimates are adjusted for stock splits as in Terry (2023), and the data come from I/B/E/S. APY is the annualized percentage yield for the eight most common deposit products on RateWatch, namely, money market accounts (MMAs), savings, checking, and 3-, 6-, 12-, 24-, and 36-month certificates of deposit (CDs). We compute these rates as bank-county-quarter-level averages of weekly rates for each bank branch operating in a county. Short-maturity products include MMAs, savings, checking, and 3-month CDs, and long-maturity products include CDs with maturities of 6, 12, 24, and 36 months. The bank balance sheet data come from the FR-Y9C quarterly consolidated reports. Deposit interest expense to total deposits is expressed in annualized percentage terms. Total deposits, interest-bearing deposits, non-interest-bearing deposits, and total assets are expressed in billions of U.S. dollars. The sample period is 2003q1 to 2018q4.

	Mean	SD	p25	p50	p75	Observations
EPS surprise (%)	1.41	7.34	-1.75	1.28	5.32	521,545
APY, all products	0.85	1.18	0.08	0.25	1.11	521,545
Short-maturity products	0.40	0.71	0.05	0.10	0.40	257,392
Long-maturity products	1.28	1.38	0.20	0.60	2.21	264,153
Dep. int/deposits	1.16	1.01	0.33	0.79	1.76	9,833
Total domestic deposits	25.04	110.83	1.34	2.94	8.45	9,849
Interest-bearing deposits	17.77	74.85	1.07	2.34	6.58	9,849
Non-interest-bearing deposits	4.66	29.17	0.12	0.36	1.25	9,849
Total assets	50.17	249.40	1.70	3.84	11.15	9,849

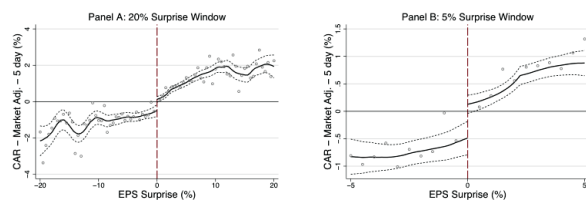


Figure 1. Return jumps around EPS surprises. In this figure, we report the market reaction to banks' earnings announcements conditional on the level of EPS surprises around zero. We measure the market reaction to earnings news using five-day buy-and-hold market-adjusted returns (BHARs) around earnings announcements, and EPS surprises as the percentage difference between realized EPS and median consensus EPS estimates between 10 and 30 weeks before the earnings announcement, as described in Section 1. In Panel A, we depict average market reactions

5.2 Figure 1: Return jumps around EPS surprises

Read: The figure shows a discontinuous announcement-return jump at zero EPS surprise in both wide and narrow windows.

Interpretation. This is the first link in the causal chain: the market penalty creates an incentive to avoid just missing.

Economic message. The return notch motivates bunching and deposit-rate cuts.

5.3 Figure 2 and Table II: Bunching and spike range

Read: Figure 2 documents abnormal mass at zero EPS surprise. Table II shows a stable spike range and a bunching lower bound around -2.5%.

Interpretation. The figure/table pair turns the visual bunching into the In Spike treatment proxy.

Economic message. Bunching provides the empirical bridge from return notches to targeting-bank observations.

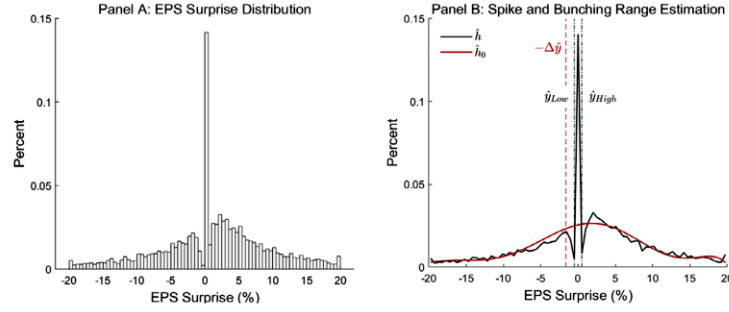


Figure 2. Bunching in bank EPS surprises. In this figure, we document bunching in the
Bunching in EPS surprises

Table II
Spike Range and Bunching Range Estimation

In this table, we report results of the spike range and bunching range estimation exercises described in Section II. In Panel A, we report estimates and semiparametrically bootstrapped standard errors for the lower bound of the spike range $\hat{y}_{Low}$, the upper bound of the spike range $\hat{y}_{High}$, and the lower bound of the bunching range $-\Delta\hat{y}$, as we vary the order k of the fitted counterfactual polynomial $\hat{h}_0$ while keeping the bin size of the realized distribution $\hat{h}$ fixed at $d = 0.1$ basis points. In Panel B, we report estimates and standard errors for the same quantities as we vary the bin size d of the realized distribution while keeping the counterfactual polynomial size fixed at $k = 7$. The EPS surprise distribution data are the same as in Figure 2. The sample period is 2003q1 to 2018q4. Standard errors (in parentheses) are computed following the semiparametric bootstrap procedure described in Section III of the Internet Appendix with 10,000 replications.

Panel A: Changing Counterfactual Polynomial Order				
	$k = 4$	$k = 5$	$k = 6$	$k = 7$ (Benchmark)
Spike lower bound, $\hat{y}_{Low}$	-0.010 (0.0031)	-0.010 (0.0032)	-0.010 (0.0032)	-0.010 (0.0033)
Spike upper bound, $\hat{y}_{High}$	0.010 (0.0034)	0.010 (0.0036)	0.010 (0.0036)	0.010 (0.0036)
Bunching range lower bound, $-\Delta\hat{y}$	-2.500 (0.1851)	-2.467 (0.2505)	-2.502 (0.2980)	-2.505 (0.7767)

Panel B: Changing Bin Size				
	$d = 10bp$	$d = 5bp$	$d = 2.5bp$	$d = 1bp$ (Benchmark)
Spike lower bound, $\hat{y}_{Low}$	-0.101 (0.0049)	-0.050 (0.0140)	-0.025 (0.0086)	-0.010 (0.0033)
Spike upper bound, $\hat{y}_{High}$	0.100 (0.0075)	0.050 (0.0132)	0.025 (0.0088)	0.010 (0.0036)
Bunching range lower bound, $-\Delta\hat{y}$	-2.261 (0.3763)	-2.398 (0.5297)	-2.468 (0.6517)	-2.505 (0.7767)

Table III
Pricing Deposits to Meet Targets

This table reports results of estimating equation (12) with incrementally restrictive combinations of fixed effects. Deposit product APYs are defined as in Table I, and *In Spike* is an indicator equal to one if the bank's quarterly EPS surprise is in the spike range and to zero otherwise. Quarter type fixed effects are equal to one if a targeting quarter follows or is followed by another targeting quarter. The sample period is 2003q1 to 2018q4. Standard errors (in parentheses) are double-clustered at the BHC and quarter level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	Dep. Variable: Product APY					
	(1)	(2)	(3)	(4)	(5)	(6)
In Spike	-0.025*** (0.01)	-0.024*** (0.01)	-0.024*** (0.01)	-0.023*** (0.01)	-0.018** (0.01)	-0.018*** (0.01)
Bank FE	Yes	No	No	No	No	No
Quarter type (QT) FE	Yes	No	No	No	No	No
Product FE	Yes	Yes	No	No	No	No
County $\times$ Year-quarter FE	Yes	Yes	Yes	Yes	No	No
Bank $\times$ QT FE	No	Yes	Yes	Yes	Yes	No
Product $\times$ County FE	No	No	Yes	Yes	No	No
Bank $\times$ Product FE	No	No	No	Yes	No	No
Bank $\times$ Product $\times$ County FE	No	No	No	No	Yes	No
Product $\times$ County $\times$ Quarter FE	No	No	No	No	Yes	Yes
Bank $\times$ QT $\times$ Product $\times$ County FE	No	No	No	No	No	Yes
R^2	0.775	0.775	0.804	0.840	0.976	0.978
Observations	521,406	521,406	520,269	520,091	307,623	306,754

To study the impact of targeting incentives on deposit quantities, we also

5.4 Table III: Pricing deposits to meet targets

Read: In Spike is negative and statistically significant across all six columns, with effects around -1.8 to -2.5 bps.

Interpretation. The effect survives increasingly saturated fixed effects, reducing concerns about bank, product, and county confounds.

Economic message. Targeting banks cut consumer deposit rates when likely trying to meet EPS targets.

5.5 Table IV: Robustness and donut tests

Read: Control-group changes preserve the negative In Spike effect; symmetric miss-or-beat donut windows do not produce similar rate cuts.

Interpretation. The effect is specific to the estimated bunching spike, not simply to being near zero EPS surprise.

Economic message. Deposit-rate cuts are tied to likely targeting behavior.

Table IV
Pricing Deposits to Meet Targets: Control Groups and Donut Tests

In this table, we ask whether our main results change depending on how we define the treated and control groups. In columns (1) and (2), we only keep control observations below and above the spike range, respectively. In column (3), we drop observations in the spike range and run a "donut" local linear regression to detect differential rate cutting by banks around (but not included in) the spike range. The low-order terms in column (3) include the percentage EPS surprise and its interaction with $I_{\epsilon \geq 0}$, an indicator for EPS surprises greater than or equal to zero. In columns (4) to (6), we drop observations in the spike range and we compare the rate-setting behavior of banks falling in a symmetric narrow range around the zero threshold to the rate-setting behavior of all other banks. This symmetric *Miss or Beat* range is $[-1\%, 1\%]$ in column (4), $[-2\%, 2\%]$ in column (5), and $[-5\%, 5\%]$ in column (6). The dependent variable in all specifications is the deposit product APY. The sample period is 2003q1 to 2018q4. Standard errors (in parentheses) are double-clustered at the BHC and quarter level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	C. Below	C. Above	Donut	Miss or Beat		
	(1)	(2)	(3)	(4)	(5)	(6)
In Spike	-0.0363*** (0.013)	-0.0192** (0.009)				
$I_{\epsilon \geq 0}$			-0.0110 (0.011)			
Miss or beat by 1%				0.0071 (0.008)		
Miss or beat by 2%					0.0001 (0.007)	
Miss or beat by 5%						0.0094 (0.007)
Low-order terms	No	No	Yes	No	No	No
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter type (QT) FE	Yes	Yes	Yes	Yes	Yes	Yes
Product FE	Yes	Yes	Yes	Yes	Yes	Yes
County $\times$ Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.782	0.776	0.775	0.775	0.775	0.775
Observations	245,662	355,731	441,344	441,344	441,344	441,344

Table V
Balance Sheet Deposit Interest Expense

In this table, we study the impact of targeting on the average rate paid on deposits at the bank level. In column (1), we use deposit interest expense normalized by total lagged deposits as the dependent variable in a bank-quarter panel. To keep estimates comparable within the table, the regression is weighted by the number of counties each bank operates in with at least one branch. In column (2), we report the same results in our main bank-county-product-quarter panel. In column (3), we study the correlation between RateWatch deposit rate quotes at the bank-county-product-quarter level (*Product APY*) and average interest paid on deposits at the bank level. In column (4), we regress the predicted value from the first-stage regression of column (3) on the *In Spike* indicator, thus studying whether the component of average bank-level interest rates explained by local quotes differs across targeting and nontargeting banks. All dependent variables are expressed as annualized percentage rates. The sample period is 2003q1 to 2018q4. Standard errors (in parentheses) are double-clustered at the BHC and quarter level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	Baseline		Prediction from Local Quotes	
	(1)	(2)	(3)	(4)
In Spike	-0.037* (0.019)	-0.033* (0.019)		-0.015*** (0.005)
Product APY			0.614*** (0.029)	
Bank FE	Yes	Yes	Yes	Yes
Product FE	No	No	Yes	Yes
QT FE	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	No	No
County $\times$ Year-quarter FE	No	No	No	Yes
R^2	0.951	0.951	0.715	0.775
Observations	9,809	517,883	517,883	521,406

5.6 Table V: Balance sheet deposit interest expense

Read: In Spike lowers bank-level deposit interest expense by roughly 3.3–3.7 bps, and product APY strongly predicts interest expense.

Interpretation. Product-level rate cuts reach the income statement and can help raise EPS.

Economic message. The table validates the mechanism behind rate-cutting.

5.7 Table VI: Monetary policy and deposit maturity

Read: Low-rate environments partially reverse the baseline effect, while short- and long-maturity products respond differently.

Interpretation. Monetary policy and deposit maturity constrain how much banks can cut rates.

Economic message. The targeting tool is state-dependent.

Table VI
Monetary Policy and Deposit Maturity

In this table, we study the interaction between targeting-induced deposit rate-cutting and monetary policy. In the first column, we augment our baseline specification (1) from Table III with the interaction between the *In Spike* indicator and a *Low Effective Fed Funds (FF) Rate* variable indicating periods of below-median levels of the effective Fed funds rate in our sample. We also include *Quarter Type* $\times$ *Low Effective FF Rate* fixed effects to control for possible interactions between monetary policy and targeting incentives over different horizons. In the second column, we replace *Low Effective FF Rate* with indicators for quarters of below-median levels of the target Fed funds rate. In columns (3) to (6), we split the sample between short- and long-maturity products. Short-maturity products include checking, savings, MMAs, and 3-month CDs. Long-maturity products include CDs with maturities of 6, 12, 24, and 36 months. Data on the Fed funds rate come from the Federal Reserve Bank of St. Louis' website. The sample period is 2003q1 to 2018q4. Standard errors (in parentheses) are double-clustered at the BHC and quarter level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	All Products		Short Maturity		Long Maturity	
	(1)	(2)	(3)	(4)	(5)	(6)
In Spike	-0.046*** (0.01)	-0.049*** (0.01)	-0.027** (0.01)	-0.029** (0.01)	-0.062*** (0.02)	-0.062*** (0.02)
In Spike $\times$ Low effective FF rate	0.042*** (0.02)		0.031** (0.01)		0.051** (0.02)	
In Spike $\times$ Low target FF rate		0.044*** (0.02)		0.033* (0.02)		0.048** (0.02)
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Product FE	Yes	Yes	Yes	Yes	Yes	Yes
County $\times$ Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
QT $\times$ Low effective FF rate FE	Yes	No	Yes	No	Yes	No
QT $\times$ Low target FF rate FE	No	Yes	No	Yes	No	Yes
R^2	0.775	0.775	0.621	0.621	0.964	0.964
Observations	521,406	521,406	256,946	256,946	264,153	264,153

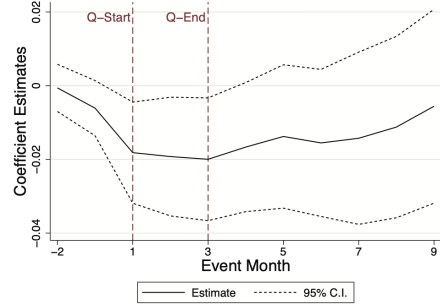


Figure 3. Deposit rate dynamics around targeting quarters. In this figure, we plot the coefficients β_2 and associated 95% standard errors (double-clustered at the cohort and calendar month levels) from estimating the stacked difference-in-differences specification described in equation (14), using event-month $Mo/E = -3$ as the baseline. In the figure, we also indicate the beginning and the end of the targeting quarter (dashed vertical lines in maroon). The sample period is 2003q1 to 2018q4.

5.8 Figure 3: Deposit-rate dynamics

Read: Rate cuts start in the targeting quarter, deepen within the quarter, and reverse afterward, with little evidence of pre-trends.

Interpretation. Timing supports a temporary targeting mechanism rather than a permanent low-rate strategy.

Economic message. The price adjustment is short-lived and centered on the reporting target.

5.9 Table VII: Deposit outflows

Read: Targeting banks lose total domestic deposits and interest-bearing deposits, while noninterest-bearing deposits do not significantly change.

Interpretation. Rate-sensitive deposits respond to lower rates, measuring the franchise-cost side of targeting.

Economic message. Meeting the target is not free.

Table VII
Deposit Outflows

In this table, we use Call Report data from the Federal Reserve to study the impact of targeting on deposit quantities. We report the results of estimating equation (13) using total domestic deposits, interest-bearing deposits, and non-interest-bearing deposits (all normalized by lagged assets) as dependent variables. As in Table V, the bank-level regressions are weighted by the number of counties a bank operates in with at least one branch. The sample period is 2003q1 to 2018q4. Standard errors (in parentheses) are double-clustered at the BHC and quarter level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	Total Domestic Deposits	Interest-Bearing Deposits	Non-Interest-Bearing Deposits
	(1)	(2)	(3)
In Spike	−0.009*** (0.003)	−0.012*** (0.004)	0.004 (0.003)
Bank FE	Yes	Yes	Yes
QT FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
R^2	0.800	0.798	0.687
Observations	9,348	9,348	9,348

Table VII using stacked difference-in-differences specifications of the form

5.10 Figure 4: EPS counterfactual

Read: Estimated rate-cutting changes EPS enough to move many targeting banks from below zero to at or above the target.

Interpretation. This directly connects deposit-rate cuts to the bunching mass in the EPS surprise distribution.

Economic message. Rate-setting is quantitatively capable of moving EPS surprises across zero.

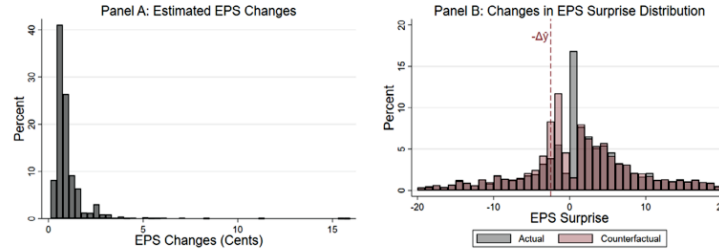


Figure 4. Rate-cutting and changes in the EPS surprise distribution. In this figure, we present the distribution of estimated EPS changes stemming from banks' rate-cutting behavior (Panel A), as well as the estimated EPS surprise distribution in a counterfactual scenario in which targeting banks do not cut rates (Panel B). The sample period is 2003q1 to 2018q4.

Rate-cutting and changes in EPS surprises

5.11 Table VIII: Local competitive spillovers

Read: Nontargeting banks raise rates when more local banks are in the spike range; private banks respond too.

Interpretation. Competitors use targeting banks' rate cuts as an opportunity to attract depositors.

Economic message. Competition partly offsets the consumer-price distortion.

5.12 Table IX: Spillover heterogeneity

Read: Spillovers vary with distance, concentration, fees, salaries, and local presence.

Table VIII
Local Competitive Spillovers

In this table, we study targeting-induced competitive spillovers on all local banks, as well as on private and public banks separately. In columns (1) and (2), we estimate equation (19) by regressing average county-product-quarter-level rates for public and private banks on the *In Spike* indicator, on the interaction between the *In Spike* indicator and the fraction of local banks that are targeting (*% In Spike*), thus capturing spillovers on other targeting banks and on the interaction between one minus the *In Spike* indicator and *% In Spike*, thus capturing spillovers on nontargeting banks. In columns (3) and (4), we separately estimate competitive spillovers on nontargeting public banks and private banks. The sample includes all public and private banks with deposit rates available on RateWatch. The sample period is 2003q1 to 2018q4. Standard errors (in parentheses) are clustered at the county level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	Dep. Variable: Average Product APY			
	(1)	(2)	(3)	(4)
In Spike	-0.034*** (0.01)	-0.034*** (0.01)	-0.033*** (0.01)	-0.033*** (0.01)
In Spike $\times$ % In Spike	0.021 (0.02)	0.022 (0.02)	0.020 (0.02)	0.022 (0.02)
(1-In Spike) $\times$ % In Spike	0.095*** (0.01)	0.095*** (0.01)		
Public $\times$ (1-In Spike) $\times$ % In Spike			0.128*** (0.03)	0.129*** (0.03)
(1-Public) $\times$ % In Spike			0.079*** (0.02)	0.078*** (0.02)
Public	-0.196*** (0.01)	-0.196*** (0.01)	-0.197*** (0.01)	-0.197*** (0.01)
Product FE	Yes	No	Yes	No
County FE	Yes	No	Yes	No
Product $\times$ County FE	No	Yes	No	Yes
Year-Quarter FE	Yes	Yes	Yes	Yes
R ²	0.794	0.807	0.794	0.807
Observations	1,546,568	1,546,286	1,546,568	1,546,286

Table IX
Competitive Spillovers in the Cross-Section

In this table, we study spillover heterogeneity based on local deposit market characteristics. In columns (1) and (2), we study branch distance and deposit market structure. In columns (3) to (5), we study characteristics of banks operating in local markets, including deposit fees, employee productivity as proxied by average salary, and local presence. *Distance* is the average physical distance (in tens of kilometers) between all bank branch pairs in the county, and *High Dep. HHI* is an indicator equal to one if local deposit HHI is above the unconditional sample median and to zero otherwise. The data come from the FDIC Summary of Deposits (SOD). *High Dep. Fees* and *High Salaries* are indicators equal to one if the deposit fees and salaries of the average bank operating in the county are above the quarterly median and to zero otherwise. The data come from the FDIC Call Reports, collapsed at the county-quarter level. *High % Local Banks* is an indicator equal to one if the deposit shares held locally by the average bank in the county is higher than the annual median and to zero otherwise. The data come from the FDIC SOD. All of the remaining variables are identical to those in Table VIII. The sample period is 2003q1 to 2018q4. Standard errors (in parentheses) are clustered at the county level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	Dep. Variable: Average Product APY				
	(1)	(2)	(3)	(4)	(5)
In Spike	-0.043*** (0.01)	-0.036*** (0.01)	-0.058*** (0.01)	-0.067*** (0.01)	-0.029** (0.01)
(1-In Spike) $\times$ % In Spike	0.066*** (0.02)	0.110*** (0.02)	0.150*** (0.03)	0.054** (0.03)	0.069*** (0.02)
(1-In Spike) $\times$ % In Spike $\times$ Distance	0.025* (0.01)				
(1-In Spike) $\times$ % In Spike $\times$ High Dep. HHI		-0.060* (0.04)			
(1-In Spike) $\times$ % In Spike $\times$ High Dep. Fees			-0.075** (0.03)		
(1-In Spike) $\times$ % In Spike $\times$ High salaries				0.056* (0.03)	
(1-In Spike) $\times$ % In Spike $\times$ High % local banks					0.079** (0.03)
Low order terms	Yes	Yes	Yes	Yes	Yes
Product $\times$ County FE	Yes	Yes	Yes	Yes	Yes
Year-Quarter FE	Yes	Yes	Yes	Yes	Yes
R ²	0.807	0.807	0.807	0.807	0.807

Interpretation. The heterogeneity supports a switching-cost interpretation and shows that competitive discipline depends on local frictions.

Economic message. Product-market structure determines how far targeting distortions spread.

6 Short summary

- Public banks face a negative return notch when they miss EPS targets.
- The EPS surprise distribution bunches just above zero.
- Likely targeting banks cut deposit rates by roughly 2 to 4 basis points.
- Rate cuts pass through to deposit interest expense and help meet EPS targets.
- Targeting banks experience deposit outflows, especially in interest-bearing deposits.
- Competitors raise deposit rates and attract some of the lost deposits.
- Competition partially checks targeting-induced price reductions.

7 Expanded conclusion

7.1 Core conclusion

The paper shows that pressure to meet quarterly EPS targets affects real consumer product pricing. In banking, the relevant product price is the deposit APY. Public banks with likely targeting incentives cut rates, reduce interest expense, and move reported EPS across the target threshold.

7.2 Full evidence chain

The evidence chain has four links: investors penalize just-missing banks; the EPS surprise distribution bunches above zero; likely targeting banks cut deposit rates during the targeting quarter; and these cuts reduce deposit expense, help explain bunching, and produce deposit outflows.

7.3 Economic mechanism

The mechanism is a short-run benefit versus long-run franchise-cost trade-off. Lowering deposit rates immediately reduces expenses and raises EPS, but it worsens the bank's value proposition to depositors and leads to outflows.

7.4 Role of competition

Competition is the paper's key product-market extension. Nontargeting banks respond by raising rates, which compensates customers for switching and lets competitors attract deposits. Thus, competition can partly check public firms' ability to pass target pressure to consumers.

7.5 What not to overinterpret

Deposit rate-setting is not the only targeting tool, and In Spike is a high-probability proxy rather than direct observation of managerial intent. The strongest interpretation is that deposit pricing is one measurable and economically important channel through which targeting pressure affects real decisions.

7.6 Broader implications

The paper extends the consequences of short-term capital-market pressure from investment distortions to product-market pricing. It suggests that quarterly targets can affect consumers, competitors, and local market shares.

7.7 Final takeaway

Meeting EPS targets is not just an accounting-market phenomenon. Capital-market incentives travel through product markets, and competitive forces determine how much of the distortion ultimately reaches consumers.